

ANKAN MANDAL

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SUMMARY

Data Science graduate student with experience in machine learning, NLP, and data-driven systems. Skilled in building predictive models, statistical analysis, and scalable data solutions using Python, SQL, and cloud technologies.

EDUCATION

Master of Science in Data Science

University of Maryland, College Park, MD, USA

Expected Graduation: May 2026

Bachelor of Technology in Computer Science and Engineering

Academy of Technology, West Bengal, India

August 2016 – June 2020

TECHNICAL SKILLS

- **Programming:** Python, SQL
- **Machine Learning & NLP:** Classification, Regression, Model Evaluation, BERT, LSTM, TF-IDF
- **Algorithms & Techniques:** PCA, Kernel PCA, Isomap, SVM, kNN, Random Forest, Logistic Regression
- **Libraries:** Pandas, NumPy, Scikit-learn, TensorFlow/Keras
- **Visualization:** Matplotlib, Seaborn
- **Tools & Databases:** MySQL, Git, Jupyter Notebook

PROJECTS

Sentiment Analysis & Hate Speech Detection | NLP, BERT, LSTM

Oct 2024 – Nov 2024

- Built an NLP pipeline to classify 25,000+ tweets into hate, offensive, and neutral categories
- Implemented TF-IDF, LSTM, and BERT; achieved ~90% accuracy using BERT
- Performed preprocessing and handled class imbalance for improved model performance
- Evaluated models using precision, recall, F1-score, and SHAP interpretability

Dimensionality Reduction & Classification on MNIST | Python, scikit-learn

Feb 2025 – Mar 2025

- Applied PCA, Kernel PCA, and Isomap on a 70K image dataset for feature reduction
- Trained SVM and kNN models; achieved ~96% accuracy using PCA + SVM
- Analyzed trade-offs between accuracy and computational cost
- Visualized embeddings to study feature representation impact

CareFlow AI (Cloud-Based Data & API System) | Python, APIs, System Design

Sep 2025 – Dec 2025

- Developed a cloud-based system integrating APIs for data processing and workflow automation
- Designed backend architecture for scalable data storage and retrieval
- Implemented system design concepts for real-world data pipelines
- Improved data accessibility and workflow efficiency

EXPERIENCE

Assistant Manager

New Mandal Hardware Stores (Authorized Dealer of TATA TISCON)

India | June 2021 – December 2023

- Managed client database, sales records, and inventory tracking systems
- Performed demand forecasting using historical data and basic statistical analysis
- Supported online sales operations and maintained accurate transaction data
- Generated structured reports for stock planning and business decision-making